



Bridging the Training–Inference Gap in TTS: Training Strategies for Robust Generative Postprocessing for Low-Resource Speakers

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Abstract

Modern text-to-speech synthesis systems usually consist of an acoustic model generating speech features, e.g., mel spectrograms, and a vocoder converting them into speech waveforms. The vocoder is typically trained with ground-truth features but receives features from the acoustic model during inference, leading to a mismatch between training and inference. To address this issue, previous work proposed employing generative postprocessing models to make the synthetic features appear more natural. While such systems can produce speech nearly indistinguishable from real speech when sufficient training data is available, their performance degrades with limited data. To mitigate this limitation, we propose a training data generation procedure using a subsampling strategy and multiple acoustic models. We evaluate it through listening tests, demonstrating consistent improvements in the naturalness of the synthetic speech across different postprocessing models and low-resource target speakers.

Index Terms: text-to-speech synthesis, data generation, generative models, low-resource speech synthesis

1. Introduction

Modern text-to-speech (TTS) synthesis systems typically comprise a neural acoustic model that converts symbolic text inputs to speech features, such as mel-scaled spectrograms, and a vocoder that transforms these speech features to speech waveforms [1]. The acoustic model is trained using corpora containing ground-truth speech features extracted from real speech recordings and their corresponding transcripts. This training usually employs a discriminative approach by minimizing a reconstruction loss, which results in the so-called “over-smoothing” effect [2], which means that the generated spectrograms lack fine-grained noise-like texture present in real spectrograms. Since vocoders are usually trained with ground-truth speech features, there is a discrepancy between training (where the vocoder is exposed to ground-truth speech features) and inference (where the vocoder receives synthetic over-smoothed speech features), which induces artifacts in the generated speech waveforms.

An early approach to circumvent this issue was to train the vocoder with synthetic spectrograms instead of real spectrograms [3]. However, this solution is expensive since it requires retraining the vocoder each time the acoustic model is updated. Since acoustic models typically train much faster than vocoders, having a universal vocoder [4] trained with real spectrograms independent of specific acoustic models is preferable.

As an alternative strategy to compensate for the over-smoothing issue, several studies investigated generative postprocessing blocks for enhancing the acoustic model output, for

example, utilizing restricted Boltzmann machines or deep belief networks [5], generative adversarial networks (GANs) [6–8], or normalizing flows [9]. These models are trained with corresponding pairs of synthetic spectrograms from the acoustic model and ground-truth spectrograms. Then, they can enhance the naturalness of the synthetic spectrograms when sufficient training data (e.g., 5 hours or more) is available for a particular speaker, referred to as a high-resource speaker in the following. However, such systems perform poorly when only limited training data (e.g., less than 15 minutes) for the target speaker is available, denoted by a low-resource speaker in the following. The decline in performance is primarily due to two reasons: First, the acoustic model produces more pronounced artifacts for the low-resource speaker, and, second, the generative postprocessing model is exposed to only a few training pairs to learn how to compensate for such strong artifacts.

Please note that the over-smoothing problem may not be as pertinent for modern fully-generative TTS systems [10, 11]. Nonetheless, traditional discriminative systems remain significant due to their superior resource efficiency, in terms of time and hardware, during training and inference. Additionally, these systems perform better with limited training data, making them preferable for applications where these aspects are critical.

In this paper, we propose a training data generation strategy to improve the performance of generative postprocessing models for low-resource speakers. Our approach increases the number of training samples, having characteristics typical of those generated by acoustic models trained on low-resource speaker data, by utilizing multiple acoustic models trained on different subsets of the training dataset. In our experiments, we apply this approach to two different generative postprocessing models, one based on GANs [8] and another on conditional flow matching (CFM) [12]. In both settings, we test both female and male low-resource target speakers. Across both generative postprocessing models and target speakers, our approach consistently yields an increased speech naturalness. Consequently, our proposed approach offers a robust improvement for low-resource TTS applications across diverse system configurations.

2. Training Data Generation

In this section, we outline the standard approach to curate the training data for generative postprocessing models in Section 2.1 and our proposed approach in Section 2.2. Both approaches support any number of high-resource single-speaker datasets in addition to a low-resource single-speaker dataset. However, for clarity, we illustrate the procedures using two high-resource single-speaker datasets.

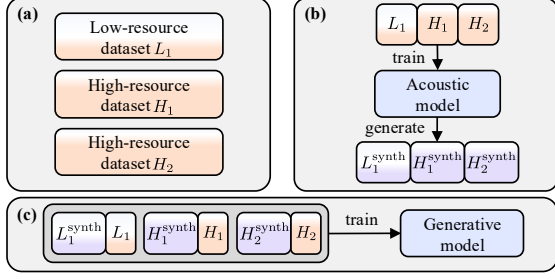


Figure 1: Overview of the standard approach to train a generative postprocessing model, comprising (a) data sources, (b) acoustic model training, and (c) generative model training.

2.1. Standard Procedure

In the standard approach, as illustrated in Figure 1, we use two high-resource single-speaker datasets H_1 and H_2 and one low-resource single-speaker dataset L_1 (Figure 1a) to train an acoustic model. Afterward, using the ground-truth transcriptions from each dataset, the acoustic model generates corresponding synthetic speech features, resulting in H_1^{synth} , H_2^{synth} , and L_1^{synth} (Figure 1b). Finally, we build pairs of corresponding real and synthetic speech samples to train the generative post-processing models (Figure 1c).

Note that the low-resource characteristic of L_1 will cause stronger artifacts in L_1^{synth} compared to the ones in H_1^{synth} and H_2^{synth} . Furthermore, there are significantly fewer samples in $(L_1^{\text{synth}}, L_1)$ than in the other pairs. Consequently, there is a high risk that the generative model may not learn well how to compensate for the stronger artifacts of the low-resource target speaker.

2.2. Proposed Procedure

In our proposed training procedure, we simulate several low-resource speakers based on the high-resource speakers to train several acoustic models, which then generate the training data for the generative postprocessing model. This approach is illustrated in Figure 2. To prepare the training data for the acoustic models, we first subsample our high-resource datasets H_1 and H_2 , yielding the simulated low-resource datasets H'_1 and H'_2 , respectively, each having approximately the same total duration as the actual low-resource dataset L_1 (Figure 2a). Then, we train three multi-speaker acoustic models A_1 , A_2 , and A_3 , using the original data for A_1 and one of the simulated low-resource datasets for A_2 and A_3 , respectively (see Figure 2b). In particular, we use L_1 , H_1 , and H_2 for A_1 ; L_1 , H'_1 , and H_2 for A_2 ; and L_1 , H_1 , and H'_2 for A_3 . Next, we use the acoustic models to generate low-quality speech features as training data for the generative model. For this purpose, we employ the corresponding acoustic model for the respective dataset that was previously used in its low-resource variant to train the model. In particular, we use A_1 to generate speech features for L_1 denoted by L_1^{synth} , A_2 for H_1 denoted by H_1^{synth} , and A_3 for H_2 denoted by H_2^{synth} . We pair the synthesized speech features $(L_1^{\text{synth}}, H_1^{\text{synth}}, \text{and } H_2^{\text{synth}})$ with the ground-truth speech features from the corresponding datasets $(L_1, H_1, \text{and } H_2)$. Then, we can train a single generative model on this compound training data, which learns to improve the acoustic model output for low-resource voices (see Figure 2c). Combining A_1 and the generative model yields an acoustic model of improved quality

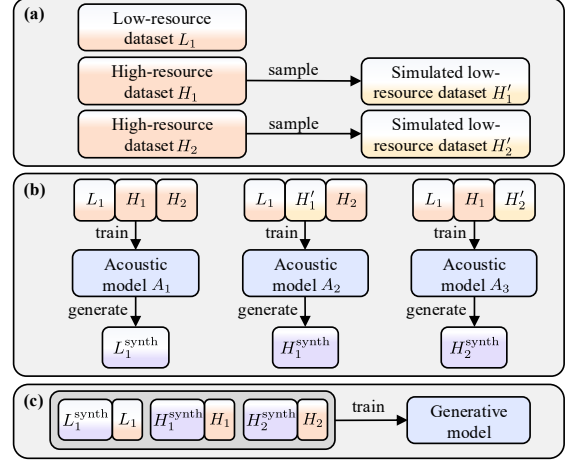


Figure 2: Overview of the proposed approach to train a generative postprocessing model, comprising (a) data sources, (b) acoustic model training, and (c) generative model training.

compared to A_1 alone.

Our approach requires that the synthetic and real examples in the training set for the generative model are as similar as possible except for their naturalness. Thus, we enforce the same prosody (pitch, speech rhythm, energy) in synthetic and real examples by employing an acoustic model with explicit prosody control [13–16], where we use ground-truth prosody features derived from the real examples to generate the corresponding synthetic examples.

Another aspect that influences the performance of our proposed approach is the sampling procedure to simulate low-resource sets from the high-resource sets. One option is to use random sampling without any restrictions. However, due to this naive approach, the durations of the sampled recordings might be significantly different from those of the actual low-resource set. Ideally, the simulated low-resource sets should be as similar as possible to the actual low-resource set. To this end, we use an improved sampling strategy, where we estimate the probability density function (PDF) of the low-resource dataset’s recording durations by applying a kernel-density estimate using Gaussian kernels. Then, we evaluate this PDF on the durations from a given high-resource dataset and use the obtained probabilities as weights in a weighted sampling procedure without replacement such that a sample cannot be drawn multiple times. The sampling process continues until the total duration of the simulated low-resource set approximates that of the actual low-resource set.

3. Experimental Setup

3.1. Training

For our experiments, we selected three speakers from the Hi-Fi TTS dataset [17] as high-resource speakers and two speakers from the PTDB-TUG dataset [18] as low-resource speakers, as listed in Table 1. Since the Hi-Fi TTS dataset provides more data than needed for the high-resource speakers, we utilized a subset of about 15 hours for each speaker. This subset was compiled as follows: We initially trained a preliminary acoustic model using all data from the three Hi-Fi TTS speakers. Then, after training, we retained only the samples with the lowest spectral reconstruction loss, which amounted to 15

Table 1: Overview of the data used in the experiments. The 30 validation samples from PTDB-TUG, which we excluded from training, are included here.

Dataset	Speaker	Gender	Duration	Samples
Hi-Fi TTS [17]	92	female	15:00:00	18 899
Hi-Fi TTS [17]	6097	male	15:00:00	19 904
Hi-Fi TTS [17]	9017	male	15:00:07	13 703
PTDB-TUG [18]	F05	female	00:13:11	221
PTDB-TUG [18]	M03	male	00:14:25	221

hours. This approach also served as a data-cleaning method, effectively excluding samples with incorrect phoneme annotations or unusual speaking styles. As for PTDB-TUG, we used all the data from the respective speakers but excluded 15 test samples per speaker from the training set.

We conducted separate experiments using either the female low-resource target speaker F05 or the male low-resource target speaker M03 (see Table 1). Each experiment involved training a multi-speaker acoustic model using four speakers, comprising three high-resource speakers from Hi-Fi TTS and one low-resource speaker from PTDB-TUG. While only two high-resource speakers were utilized for the illustrative description in Section 2, our experiments incorporated three high-resource speakers, leading to four acoustic models in the proposed methodology.

The employed acoustic model was an extended version of Forward Tacotron, as described in prior work [16]. This model features a parallel neural architecture, uses text in phoneme representation (IPA) and a speaker identifier [19] as input, and predicts a mel-scaled spectrogram with logarithmic magnitude compression. The mel spectrograms in our experiments were computed from speech recordings with a consistent sampling rate of 22 050 Hz, using a blocksize of 46.4 ms, a hop size of 11.6 ms, and 80 mel bands with center frequencies spread between 0 and 8 kHz). The model includes semantically meaningful variation predictors for pitch, energy, voicing confidence, and phoneme durations to model prosody, similar to FastTacotron [13]. The model was trained for 200K training iterations using the Adam optimizer [20] with an initial learning rate of 10^{-3} for the first 50K iterations and 10^{-4} afterwards. In our experiments, we utilized a pre-trained version of StyleMelGAN [21] as the vocoder.

3.2. Models Under Test

As a baseline, the mel spectrograms from the standard Forward Tacotron version were directly passed to the vocoder. Another baseline involved training Forward Tacotron with noise augmentation for the low-resource speaker, a method proven to enhance the TTS quality in low-resource scenarios [22]. For this variant, we chose a slightly different learning rate schedule, which yielded better quality in preliminary experiments, starting with a learning rate of 10^{-4} for the first 150K iterations and 10^{-5} thereafter.

The first generative postprocessing module utilized was a GAN-based system, as described in previous work [8]. This system uses a generator with a convolutional U-Net architecture, similar to the one in the pix2pix system [23], conditioned on the spectrogram generated by the acoustic model.

The second generative model employed was a CFM-based system known as FlowMAC [12]. Unlike the original model used for audio coding, we only utilized its CFM module, omitting its encoder-decoder frontend. This CFM module is a

Transformer-based U-Net, also conditioned on the spectrogram from the acoustic model. Furthermore, we adopted the same normalization process as that used for the GAN [8], normalizing each mel band to a mean of zero and a standard deviation of one for both the conditioning and target mel spectrogram separately, using statistics estimated over the training set. The remaining hyperparameters for both generative models were consistent with those specified in the original publications [8, 12], including the inference setup of the CFM-based model, using 32 steps for the ODE solver, applying the Euler method with a factor of one for the classifier-free guidance. Several studies [24, 25] have reported improved inference performance with the introduction of a temperature parameter τ , which scales the prior distribution. We experimented with various settings for this parameter and determined that $\tau = 0.7$ yielded the best average objective metric (described below) for a subset of 50 samples from the training data. We tested both generative models using the standard and our proposed training data generation procedure (outlined in Section 2).

3.3. Evaluation

To objectively evaluate the effectiveness of the different approaches, we apply the SCOREQ system [26] using the model for the synthetic domain and the reference mode. This system has been trained to predict the distance in mean opinion score between a reference speech signal and a test speech signal.

Additionally, we conducted a subjective multi-stimulus test (MST) with a hidden reference similar to MUSHRA [27], for each of our target speakers. Designed originally to compare audio codecs, multi-stimuli tests with hidden references are also commonly utilized to compare TTS systems [21, 28–32]. In the MST, the participants were provided with a reference signal derived by vocoding the ground-truth mel spectrogram from a hold-out test sample. Then, they were asked to rate the overall quality of several synthetic versions of the same utterance on a discrete scale from 0 to 100. Among these versions, a hidden reference signal was included, and the participants were instructed to assign it a score of 100. In total, 41 participants took part in the test for speaker F05, while 37 participants assessed the test for speaker M03, each rating seven different synthetic renditions of twelve distinct texts. For our evaluation, we excluded participants who rated the hidden reference with a score below 90 more than twice out of twelve instances, resulting in 28 and 23 participants retained for the tests of speakers F05 and M03, respectively.¹

In our informal listening impressions, we occasionally noted that the best synthetic samples unexpectedly appeared superior to the references, which can not be adequately addressed in MUSHRA-like tests, where the reference is assumed to represent the highest quality. To further investigate potential improvements over the reference, we also conducted a ranking-by-elimination (RBE) test [33]. In the RBE test, the participants were presented with the same synthetic samples as in the MST but without an open reference. Their task was to identify the item with the lowest overall quality among all items, which was then removed, and to repeat this procedure until no items were left or until all remaining items were perceived to be of the same quality. To analyze the ratings obtained from our participants, we employed the Plackett-Luce model [34, 35] to convert the rankings into scores (or so-called worth values). A total of 15 participants took part in the test for speaker F05, while 16

¹Samples available: <https://www.audiolabs-erlangen.de/resources/NLUI/2025-Interspeech-LowResGen>

Table 2: Mean and 95 % confidence intervals of different scores yielded for the approaches under test.

Condition	SCOREQ (↓)			F05	MST (↑)			F05	RBE (↑)		
	F05	M03	∅		F05	M03	∅		F05	M03	∅
Reference	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	98.4 ± 0.7	98.2 ± 0.7	98.3 ± 0.5		0.0 ± 0.2	0.0 ± 0.2	0.0 ± 0.2	
Forward_Tac	0.37 ± 0.04	0.42 ± 0.03	0.39 ± 0.03	50.2 ± 3.0	51.8 ± 3.0	50.9 ± 2.1		-3.1 ± 0.3	-3.2 ± 0.3	-3.1 ± 0.2	
Noise_Aug	0.45 ± 0.08	0.41 ± 0.06	0.43 ± 0.05	44.1 ± 3.1	48.1 ± 3.1	45.9 ± 2.2		-3.6 ± 0.3	-3.6 ± 0.3	-3.6 ± 0.2	
GAN_Standard	0.33 ± 0.04	0.34 ± 0.03	0.34 ± 0.03	67.2 ± 2.7	78.6 ± 2.2	72.3 ± 1.8		-1.2 ± 0.2	-0.9 ± 0.2	-1.1 ± 0.1	
GAN_Proposed	0.31 ± 0.03	0.34 ± 0.04	0.32 ± 0.02	71.6 ± 2.5	78.7 ± 2.3	74.8 ± 1.7		-0.7 ± 0.2	-0.8 ± 0.2	-0.7 ± 0.1	
CFM_Standard	0.25 ± 0.02	0.30 ± 0.03	0.28 ± 0.02	75.4 ± 2.6	83.4 ± 2.0	79.0 ± 1.7		-0.7 ± 0.2	-0.6 ± 0.2	-0.6 ± 0.1	
CFM_Proposed	0.24 ± 0.02	0.30 ± 0.03	0.27 ± 0.02	76.2 ± 2.4	84.0 ± 2.1	79.8 ± 1.7		-0.5 ± 0.2	-0.5 ± 0.2	-0.5 ± 0.1	

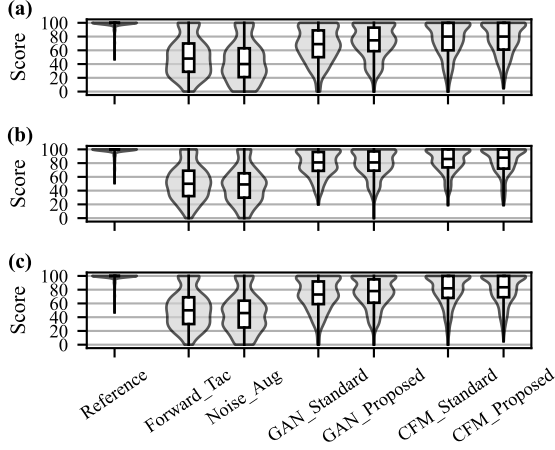


Figure 3: Violin plots for the MST scores for speaker (a) F05, (b) M03, and (c) both combined. The violin plots include boxes that indicate the first, second, and third quartiles.

participants assessed the test for speaker M03, with each participant again rating seven different synthetic renditions of twelve distinct utterances. All participants in the MST and RBE tests were experienced with listening tests.

Our method aims to enhance the naturalness of the synthetic audio quality, not the prosody. Therefore, we conditioned the acoustic model on ground-truth prosody features in both objective and subjective evaluations.

4. Results

The left columns of Table 2 present statistics of the SCOREQ results for the 15 hold-out test samples. The reference was obtained by passing the ground-truth spectrograms to the vocoder. Since SCOREQ measures the perceptual distance to this reference, the reference itself naturally yields a distance of 0.0. The Forward Tacotron baseline (Forward_Tac) achieved an average distance of 0.39. Interestingly, the noise augmentation strategy (Noise_Aug) did not lead to an improvement, yielding an average distance of 0.43. Preliminary investigations suggest that the noise augmentation strategy primarily benefits prosody prediction rather than audio naturalness, which was the focus of our evaluation. Since we used ground-truth prosody features, this could potentially explain its lower performance in this context. In contrast, the postprocessing GAN strongly improved the performance, achieving an average distance of 0.34 with the standard approach (GAN_Standard) and 0.32 with the proposed approach (GAN_Proposed). The CFM-based postprocessing model further reduced the average distances to 0.28 and 0.27 for the standard (CFM_Standard) and proposed approaches

(CFM_Proposed), respectively. It is important to interpret these objective scores with caution, as they serve only as estimates for human ratings.

Figure 3 displays several violin plots that visualize the score distributions obtained in our MST, including boxes that indicate the first, second, and third quartiles. Figures 3a, 3b, and 3c show the results for speakers F05, M03, and both combined, respectively. Additionally, the middle columns of Table 2 present statistics of the MST scores. Similar to the objective scores, the reference yielded the highest average score of 98.3. The Forward Tacotron baseline scored lower at 50.9, while the noise augmentation strategy resulted in a lower score of 45.9. The standard and proposed GAN models yielded average scores of 72.3 and 74.8, respectively. The standard and proposed CFM models yielded average scores of 79.0 and 79.8, respectively. It is noteworthy that our proposed approach yields improvements for both speakers and for both generative models. These improvements are statistically significant for the GAN and not significant for the CFM, as determined by a Wilcoxon signed-rank test, using a significance threshold of $p \leq 0.01$.

The right columns of Table 2 present statistics of the log-worth values from the Plackett-Luce model, summarizing the ratings of our RBE test. The reference, corresponding to a worth value of 0.0, calibrates the worth values for the other conditions. Therefore, a positive worth value indicates a higher preference than the reference, while negative values indicate a lower preference. Many of the tendencies of the MST are reflected here as well. However, it is noteworthy the CFM model trained with our proposed approach obtains a high worth value of -0.5 for both speakers. This can be attributed to the fact that it was preferred over the reference in 26.6 % of the rankings obtained in the RBE test, while the reference was preferred in 50.3 % of the cases, and there was no preference in the remaining rankings.

5. Conclusion and Outlook

In this paper, we introduced a novel strategy to generate training data for generative postprocessing models in TTS systems, aiming to enhance the naturalness of synthetic speech features for low-resource speakers. The results from our listening tests demonstrate that our method is robust and consistently improves the perceived speech naturalness across different generative postprocessing model types and low-resource speaker scenarios, making it a beneficial addition to various TTS setups.

For future work, the proposed method could be extended to scenarios where many high-resource single-speaker datasets are available, but only a few are intended to be used in the training procedure. In such cases, it could be an option to select the speakers most similar to the low-resource target speaker, e.g., based on some speaker verification embedding known to capture speaker similarity [36].

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7. References

- [1] X. Tan, T. Qin, F. K. Soong, and T. Liu, “A survey on neural speech synthesis,” *CoRR*, vol. abs/2106.15561, 2021.
- [2] Y. Ren, X. Tan, T. Qin, Z. Zhao, and T. Liu, “Revisiting over-smoothness in text to speech,” in *Proc. of the Ann. Meeting of the Association for Computational Linguistics*, Dublin, Ireland, 2022, pp. 8197–8213.
- [3] J. Shen, R. Pang, R. J. Weiss, M. Schuster, N. Jaitly, Z. Yang, Z. Chen, Y. Zhang, Y. Wang, R. Ryan, R. A. Saurous, Y. Agiomyriannakis, and Y. Wu, “Natural TTS synthesis by conditioning WaveNet on mel spectrogram predictions,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Calgary, Canada, 2018, pp. 4779–4783.
- [4] S. Lee, W. Ping, B. Ginsburg, B. Catanzaro, and S. Yoon, “BigVGAN: A universal neural vocoder with large-scale training,” in *Proc. of the Int. Conf. on Learning Representations (ICLR)*, Kigali, Rwanda, 2023.
- [5] L. Chen, T. Raitio, C. Valentini-Botinhao, Z. Ling, and J. Yamagishi, “A deep generative architecture for postfiltering in statistical parametric speech synthesis,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 23, no. 11, pp. 2003–2014, 2015.
- [6] L. Sheng and E. N. Pavlovskiy, “Reducing over-smoothness in speech synthesis using generative adversarial networks,” in *Proc. of the Int. Multi-Conf. on Engineering, Computer and Information Sciences (SIBIRCON)*, Novosibirsk, Russia, 2019, pp. 972–974.
- [7] P. Neekhara, C. Donahue, M. S. Puckette, S. Dubnov, and J. J. McAuley, “Expediting TTS synthesis with adversarial vocoding,” in *Proc. of the Ann. Conf. of the Int. Speech Communication Association (Interspeech)*, Graz, Austria, 2019, pp. 186–190.
- [8] P. Sani, J. Bauer, F. Zalkow, E. A. P. Habets, and C. Dittmar, “Improving the naturalness of synthesized spectrograms for TTS using GAN-based post-processing,” in *Proc. of the ITG Conf. on Speech Communication*, Aachen, Germany, 2023, pp. 270–274.
- [9] F. Lux, J. Koch, S. Meyer, T. Bott, N. Schaffler, P. Denisov, A. Schweitzer, and N. T. Vu, “The IMS Toucan system for the Blizzard Challenge 2023,” in *Proc. of the Blizzard Challenge Workshop*, Grenoble, France, 2023, pp. 40–45.
- [10] S. E. Eskimez, X. Wang, M. Thakker, C. Li, C. Tsai, Z. Xiao, H. Yang, Z. Zhu, M. Tang, X. Tan, Y. Liu, S. Zhao, and N. Kanda, “E2 TTS: Embarrassingly easy fully non-autoregressive zero-shot TTS,” in *Proc. of the IEEE Spoken Language Technology Workshop (SLT)*, Macao, China, 2024, pp. 682–689.
- [11] E. Casanova, K. Davis, E. Gölgel, G. Gökner, I. Gulea, L. Hart, A. Aljafari, J. Meyer, R. Morais, S. Olayemi, and J. Weber, “XTTS: A massively multilingual zero-shot text-to-speech model,” in *Proc. of the Ann. Conf. of the Int. Speech Communication Association (Interspeech)*, Kos Island, Greece, 2024, pp. 4978–4982.
- [12] N. Pia, M. Strauss, M. Multus, and B. Edler, “FlowMAC: Conditional flow matching for audio coding at low bit rates,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Hyderabad, India, 2025.
- [13] D. V. Sang and L. X. Thu, “FastTacotron: A fast, robust and controllable method for speech synthesis,” in *Proc. of the Int. Conf. on Multimedia Analysis and Pattern Recognition (MAPR)*, Hanoi, Vietnam, 2021.
- [14] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T. Liu, “FastSpeech 2: Fast and high-quality end-to-end text to speech,” in *Proc. of the Int. Conf. on Learning Representations (ICLR)*, virtual, Austria, 2021.
- [15] A. Łancucki, “FastPitch: Parallel text-to-speech with pitch prediction,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Toronto, Canada, 2021, pp. 6588–6592.
- [16] F. Zalkow, P. Sani, M. Fast, J. Bauer, M. Joshaghani, K. Kayyar, E. A. P. Habets, and C. Dittmar, “The AudioLabs system for the Blizzard Challenge 2023,” in *Proc. of the Blizzard Challenge Workshop*, Grenoble, France, 2023, pp. 63–68.
- [17] E. Bakhturina, V. Lavrukhin, B. Ginsburg, and Y. Zhang, “Hi-fi multi-speaker English TTS dataset,” in *Proc. of the Ann. Conf. of the Int. Speech Communication Association (Interspeech)*, Brno, Czech Republic, 2021, pp. 2776–2780.
- [18] G. Pirker, M. Wohlmayr, S. Petrik, and F. Pernkopf, “A pitch tracking corpus with evaluation on multipitch tracking scenario,” in *Proc. of the Ann. Conf. of the Int. Speech Communication Association (Interspeech)*, Florence, Italy, 2011, pp. 1509–1512.
- [19] K. Kayyar, C. Dittmar, N. Pia, and E. A. P. Habets, “Multi-speaker text-to-speech using ForwardTacotron with improved duration prediction,” in *Proc. of the ITG Conf. on Speech Communication*, Aachen, Germany, 2023, pp. 186–190.
- [20] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. of the Int. Conf. on Learning Representations (ICLR)*, San Diego, USA, 2015.
- [21] A. Mustafa, N. Pia, and G. Fuchs, “StyleMelGAN: An efficient high-fidelity adversarial vocoder with temporal adaptive normalization,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Toronto, Canada, 2021, pp. 6034–6038.
- [22] K. K. Lakshminarayana, F. Zalkow, C. Dittmar, N. Pia, and E. A. P. Habets, “Low-resource text-to-speech synthesis using noise-augmented training of ForwardTacotron,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Hyderabad, India, 2025.
- [23] P. Isola, J. Zhu, T. Zhou, and A. A. Efros, “Image-to-image translation with conditional adversarial networks,” in *IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017, pp. 5967–5976.
- [24] S. Mehta, R. Tu, J. Beskow, É. Székely, and G. E. Henter, “Matcha-TTS: A fast TTS architecture with conditional flow matching,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Seoul, Korea, 2024, pp. 11 341–11 345.
- [25] V. Popov, I. Vovk, V. Gogoryan, T. Sadekova, and M. A. Kudinov, “Grad-TTS: A diffusion probabilistic model for text-to-speech,” in *Proc. of the Int. Conf. on Machine Learning (ICML)*, virtual, 2021, pp. 8599–8608.
- [26] A. Ragano, J. Skoglund, and A. Hines, “SCOREQ: Speech quality assessment with contrastive regression,” in *Proc. of the Ann. Conf. on Neural Information Processing Systems (NeurIPS)*, Vancouver, Canada, 2024, pp. 105 702–105 729.
- [27] International Telecommunications Union, “Recommendation ITU-R BS.1534-3: Method for the subjective assessment of intermediate quality level of audio systems,” Geneva, Switzerland, recommendation, 2015.
- [28] T. Merritt, B. Putrycz, A. Nadolski, T. Ye, D. Korzekwa, W. Dolecki, T. Drugman, V. Klimkov, A. Moinet, A. Breen, R. Kuklinski, N. Strom, and R. Barra-Chicote, “Comprehensive evaluation of statistical speech waveform synthesis,” in *Proc. of the IEEE Spoken Language Technology Workshop (SLT)*, Athens, Greece, 2018, pp. 325–331.
- [29] P. Wagner, J. Beskow, S. Betz, J. Edlund, J. Gustafson, G. Eje Henter, S. Le Maguer, Z. Malisz, É. Székely, C. Tännander, and J. Voße, “Speech synthesis evaluation — state-of-the-art assessment and suggestion for a novel research program,” in *Proc. of the ISCA Workshop on Speech Synthesis (SSW)*, Vienna, Austria, 2019, pp. 105–110.
- [30] J. Latorre, J. Lachowicz, J. Lorenzo-Trueba, T. Merritt, T. Drugman, S. Ronanki, and V. Klimkov, “Effect of data reduction on sequence-to-sequence neural TTS,” in *Proc. of the IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Brighton, United Kingdom, 2019, pp. 7075–7079.
- [31] P. Govalkar, J. Fischer, F. Zalkow, and C. Dittmar, “A comparison of recent neural vocoders for speech signal reconstruction,” in *Proc. of the ISCA Workshop on Speech Synthesis (SSW)*, Vienna, Austria, 2019, pp. 7–12.
- [32] O. Perrotin, B. Stephenson, S. Gerber, G. Bailly, and S. King, “Refining the evaluation of speech synthesis: A summary of the Blizzard Challenge 2023,” *Computer Speech & Language*, vol. 90, p. 101747, 2025.
- [33] K. Kayyar, C. Dittmar, N. Pia, and E. A. P. Habets, “Subjective evaluation of text-to-speech models: Comparing absolute category rating and ranking by elimination tests,” in *Proc. of the ISCA Workshop on Speech Synthesis (SSW)*, Grenoble, France, 2023, pp. 191–196.
- [34] R. L. Plackett, “The analysis of permutations,” *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, vol. 24, no. 2, pp. 193–202, 1975.
- [35] H. L. Turner, J. van Etten, D. Firth, and I. Kosmidis, “Modelling rankings in R: The PlackettLuce package,” *Computational Statistics*, vol. 35, no. 3, pp. 1027–1057, 2020.
- [36] B. Desplanques, J. Thienpondt, and K. Demuyne, “ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification,” in *Proc. of the Ann. Conf. of the Int. Speech Communication Association (Interspeech)*, Shanghai, China, 2020, pp. 3830–3834.